Working_with_Data

Table of contents

1 ——————Packages & library——————

```r
install.packages("car")
```

```
Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.4'
(as 'lib' is unspecified)
```

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(descr)
library(knitr)
library(dplyr)
library(Hmisc)
```

```
Attaching package: 'Hmisc'

The following objects are masked from 'package:dplyr':

    src, summarize

The following objects are masked from 'package:base':

    format.pval, units
```

```r
library(readr)
library(readxl)
library(ggplot2)
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

2 ——————————Load——————————

```r
Project_Data = read.csv("/cloud/project/Data/Connection_to_Nature_Data.csv",
header = TRUE)
```

3 ——————————Variables—————————

```r
# VARIABLE 1: People's Age


# Age 18 and above

Project_Data <- subset(Project_Data, D_Age >= 18)

# Group D_Age into 4 age range's and Labeling

Project_Data$Age_Group <- cut(Project_Data$D_Age,
                              breaks = c(18, 25, 40, 65, Inf),
                              labels = c("18-25", "26-40", "40-65", "65+"),
                              right = TRUE,
                              include.lowest = TRUE)

# Frequency table

freq(as.ordered(Project_Data$Age_Group),plot =FALSE)
```

```
as.ordered(Project_Data$Age_Group)
      Frequency Percent Cum Percent
18-25        49   9.515       9.515
26-40        85  16.505      26.019
40-65       301  58.447      84.466
65+          80  15.534     100.000
Total       515 100.000
```

I choose this variable (age) because I think it would be important to look at in reference to how loneliness and time spent in nature varies among age groups. Perhaps depending on the age group, there will be more positive benefits to those exposed to nature in relation to loneliness.

```r
# VARIABLE 2: How many hours on average do you currently spend in nature per week?

# Group D_hours into 4 categories & Labeling

Project_Data$Nature_Hours_Group <- cut(Project_Data$D_hours,
                                       breaks = c(0, 5, 15, 30, Inf),
```

```r
                                          labels = c("Low (0-5)", "Moderate (6-15)", "High (16-30)", "Very High (30+)"),
                                          include.lowest = TRUE)

# Frequency table

freq(as.ordered(Project_Data$Nature_Hours_Group), plot = FALSE)
```

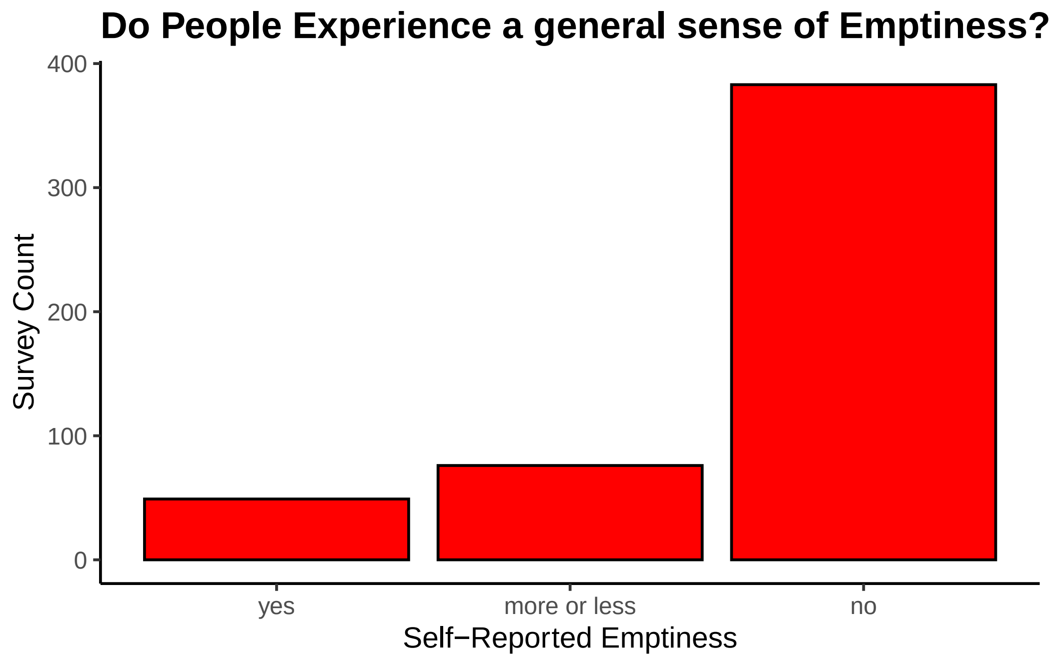
```
as.ordered(Project_Data$Nature_Hours_Group)
                Frequency Percent Cum Percent
Low (0-5)             137  26.602       26.60
Moderate (6-15)       230  44.660       71.26
High (16-30)          118  22.913       94.17
Very High (30+)        30   5.825      100.00
Total                 515 100.000
```

This is very important. This variable (hours spent in nature) is important because when I did the literature review assignment, depending on the time spent in nature, actually lowered both social loneliness and emotional loneliness, but it depends how much time was spent in nature.

```r
# VARIABLE 3: People's experience a general sense of emptiness (survey response)

# Labeling

Project_Data$Lon_1 <- factor(Project_Data$Lon_1,
                             levels = c(1, 2, 3),
                             labels = c("yes", "more or less", "no"))

# Frequency table

freq(as.ordered(Project_Data$Lon_1), plot = FALSE)
```

```
as.ordered(Project_Data$Lon_1)
             Frequency Percent Valid Percent Cum Percent
yes                 49   9.515         9.646       9.646
more or less        76  14.757        14.961      24.606
no                 383  74.369        75.394     100.000
NA's                 7   1.359
Total              515 100.000       100.000
```

```r
# Bar graph

ggplot(data = subset(Project_Data, !is.na(Lon_1)), aes(x = Lon_1)) +
  geom_bar(fill = "red", color = "black") +
  xlab("Self-Reported Emptiness") +
  ylab("Survey Count") +
  ggtitle("Do People Experience a general sense of Emptiness?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```

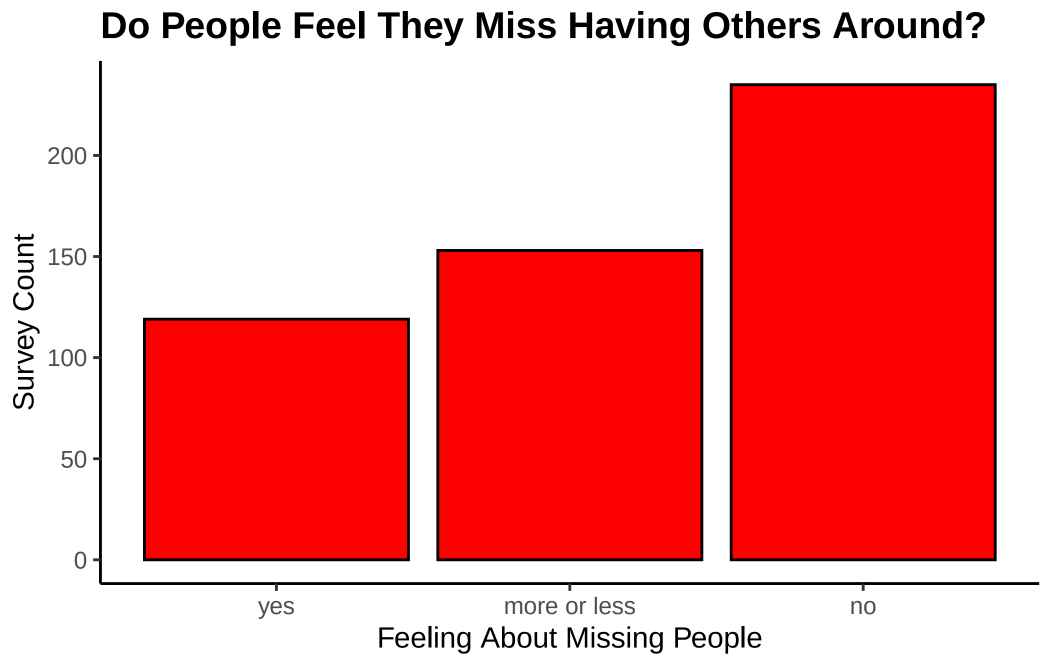


- Lit Review Assign: I choose this variable (people's sense of emptiness) because this can be a reason for social/emotional loneliness. If time spent nature is associated with lower loneliness on these two paths, we might also see a decrease in emptiness to those who spend more time in nature.
- Univariate Data Visualization Assign: The first graph illustrates people's experiences of a general sense of emptiness. Overall, most respondents said no, while the fewest number of respondents said yes.

```
# VARIABLE 4: I miss having people around (survey response)

# Labeling

Project_Data$Lon_4 <- factor(Project_Data$Lon_4,
                             levels = c(1, 2, 3),
                             labels = c("yes", "more or less", "no"))

# Frequency table

freq(as.ordered(Project_Data$Lon_4), plot = FALSE)
```

```
as.ordered(Project_Data$Lon_4)
             Frequency Percent Valid Percent Cum Percent
yes                119  23.107         23.47       23.47
more or less       153  29.709         30.18       53.65
no                 235  45.631         46.35      100.00
NA's                 8   1.553
Total              515 100.000        100.00
```

```
# Bar graph

ggplot(data = subset(Project_Data, !is.na(Lon_4)), aes(x = Lon_4)) +
  geom_bar(fill = "red", color = "black") +
  xlab("Feeling About Missing People") +
  ylab("Survey Count") +
  ggtitle("Do People Feel They Miss Having Others Around?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```



- Lit Review Assign: This variable (missing social interaction) could be important because social loneliness is being examined here. Comparing this to time spent in nature can help show whether nature can also regulate/help social loneliness as well.
- Univariate Data Visualization Assign: The second graph illustrates people's survey responses to whether individuals feel that they miss others in their lives. This is similar to the first graph. Most respondents said no, while the fewest number of respondents said yes.

```
# VARIABLE 5: I have high self-esteem (survey response)

# Labeling

Project_Data$SE_1 <- factor(Project_Data$SE_1,
                            levels = c(1, 2, 3, 4, 5),
                            labels = c("not very true of me", "2", "3", "4", "very true of me"),
                            ordered = TRUE)

# Frequency table

freq(as.ordered(Project_Data$SE_1), plot = FALSE)
```

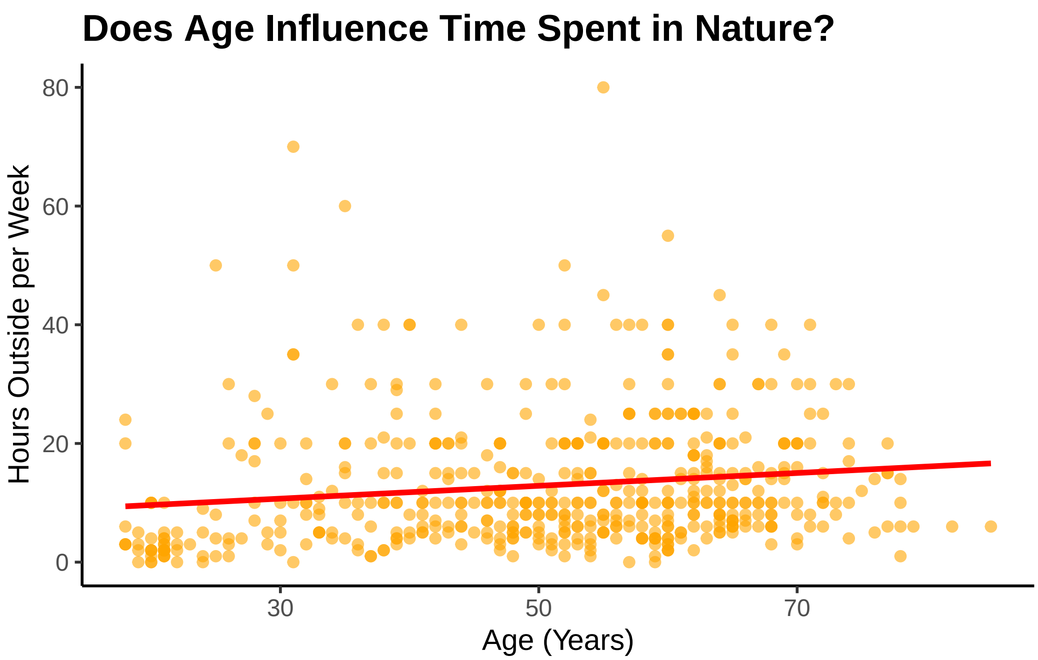
```
as.ordered(Project_Data$SE_1)
                    Frequency Percent Cum Percent
not very true of me        35   6.796       6.796
2                          69  13.398      20.194
3                         146  28.350      48.544
4                         193  37.476      86.019
very true of me            72  13.981     100.000
Total                     515 100.000
```

Lastly, I also choose this variable (people's self-esteem) because those who experience loneliness and spend little time in nature differ from those who don't feel loneliness and do spend time in nature. Perhaps those who do spend more time have higher level's of agreement to self-esteem compared to those who do not.

4 ————————Bivariate Analysis————————

```r
# Bivariate 1: Correlation between age and time spent in nature

Project_Data <- subset(Project_Data, !is.na(D_Age) & !is.na(D_hours))

# Scatter plot

ggplot(Project_Data, aes(x = D_Age, y = D_hours)) +
  geom_point(color = "orange", alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  xlab("Age (Years)") +
  ylab("Hours Outside per Week") +
  ggtitle("Does Age Influence Time Spent in Nature?") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"))
```

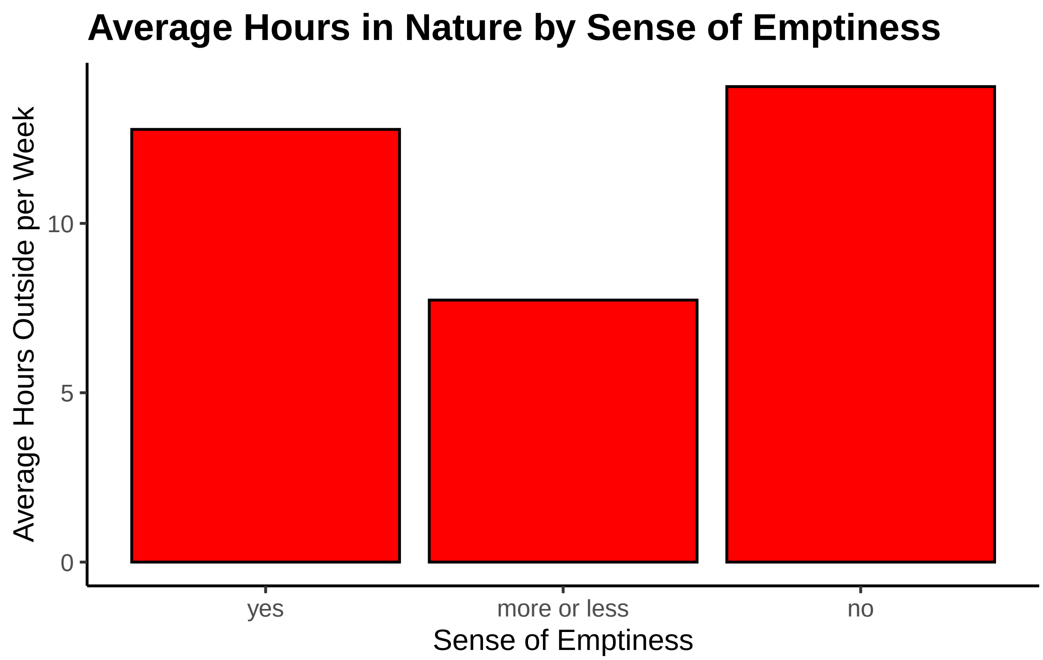
```
`geom_smooth()` using formula = 'y ~ x'
```



- The scatter plot shown here shows that there is a slight positive correlation between age and time spent outside hourly per week. The positive correlation shows that older individuals might spend a little bit more time in nature. However, when it comes to my research interest concerning the relationship between loneliness and nature, age does not seem to be a strong factor.

```r
# Bivariate 2: Correlation between time spent in nature and general emptiness

summary_data <- Project_Data %>%
  filter(!is.na(Lon_1) & !is.na(D_hours)) %>%
  group_by(Lon_1) %>%
  summarise(Avg_Hours = mean(D_hours))

# Bar graph
ggplot(summary_data, aes(x = Lon_1, y = Avg_Hours, fill = Lon_1)) +
  geom_bar(stat = "identity", fill = "red", color = "black") +
  xlab("Sense of Emptiness") +
  ylab("Average Hours Outside per Week") +
  ggtitle("Average Hours in Nature by Sense of Emptiness") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold")) +
  guides(fill = FALSE)
```

```
Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as
of ggplot2 3.3.4.
```



- The bar graph shown here illustrates the relationship between people's sense of emptiness, and their average hours spent in nature. Individuals who say YES to feeling empty, show, a high number of hours outside, and are like those who say NO to feeling empty as well. However, people who said MORE OR LESS, spent the least amount of time outside. They would mean that the relationship between a sense of emptiness and time spent in nature is not straightforward and there could be other factors influencing nature and loneliness.

5 ——————————Chi-Square Test of Independence——————————

```r
# Recode D_Gender into a factor with labels "man" and "woman"
Project_Data$D_Gender <- factor(Project_Data$D_Gender, levels = c(1, 2), labels = c("man", "woman"))

# Create a frequency table for D_Gender
gender_counts <- table(Project_Data$D_Gender)

# Print the counts
print(gender_counts)
```

```

  man woman 
  143   370 
```

```r
# Men Chi-Square
Men_Data <- subset(Project_Data, D_Gender == "man")

# Remove missing data
Men_Data_Clean <- subset(Men_Data, !is.na(Nature_Hours_Group) & !is.na(Lon_1))

# Create the contingency table
table_men <- table(Men_Data_Clean$Nature_Hours_Group, Men_Data_Clean$Lon_1)

cat("Contingency table for men:\n")
```

```
Contingency table for men:
```

```r
print(table_men)
```

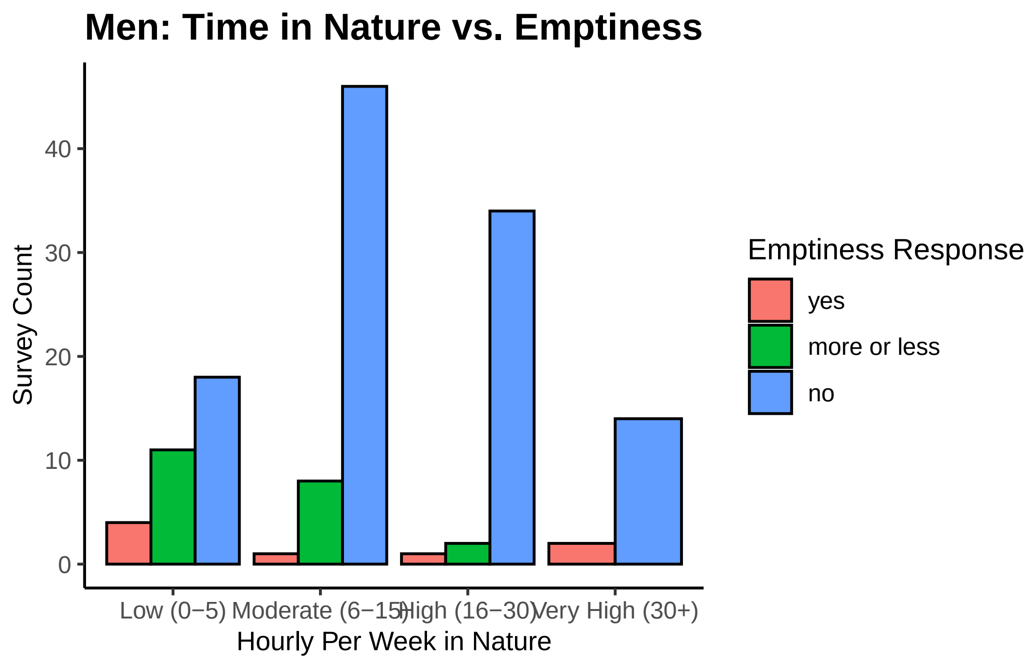
```
                  yes more or less no
  Low (0-5)         4           11 18
  Moderate (6-15)   1            8 46
  High (16-30)      1            2 34
  Very High (30+)   2            0 14
```

```r
# Chi-Square Test
if (sum(table_men) == 0) {cat("No valid data in table for men.\n")
  } else {
  chisq_test_men <- chisq.test(table_men)
  cat("\n Chi-Square Test for Men:\n")
  print(chisq_test_men)}
```

```
Warning in chisq.test(table_men): Chi-squared approximation may be incorrect
```

```

 Chi-Square Test for Men:

    Pearson's Chi-squared test

data:  table_men
X-squared = 21.456, df = 6, p-value = 0.001519
```

```r
# Bar Chart
ggplot(Men_Data_Clean, aes(x = Nature_Hours_Group, fill = Lon_1)) +
  geom_bar(position = "dodge", color = "black") +
  labs(title = "Men: Time in Nature vs. Emptiness",
       x = "Hourly Per Week in Nature",
       y = "Survey Count",
       fill = "Emptiness Response") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"),
    axis.title.x = element_text(size = 10),
    axis.title.y = element_text(size = 10),
    axis.text.x = element_text(size = 9),
    axis.text.y = element_text(size = 9))
```



Chi-Square Test of Independence Analysis #1: Men

This analysis tested whether there is a relationship between men's time spent in nature hourly per-week (grouped into four categorical levels), and their reported sense of emptiness ("yes," "more or less," and "no").

Analyzing the results, most men across all four categories reported not feeling empty. This shows a potential trend: the more time men spend time in nature each week, the less likely they are to report feelings of emptiness. However, in the "Very High (30+)" hours group, there was a modest increase in feelings of emptiness, showing that this relationship is not entirely linear and may be influenced by other factors.

The chi-square test resulted in a p-value of 0.001, making it statistically significant. This value is below the threshold of 0.05, therefore we reject the null hypothesis. Our null hypothesis was that there is no relationship between time spent in nature and symptoms of emptiness among men.

```r
# Women Chi-Square
Women_Data <- subset(Project_Data, D_Gender == "woman")

# Remove missing data
Women_Data_Clean <- subset(Women_Data, !is.na(Nature_Hours_Group) & !is.na(Lon_1))

# Contingency Table
table_women <- table(Women_Data_Clean$Nature_Hours_Group, Women_Data_Clean$Lon_1)

cat("Contingency table for women:\n")
```

```
Contingency table for women:
```

```r
print(table_women)
```

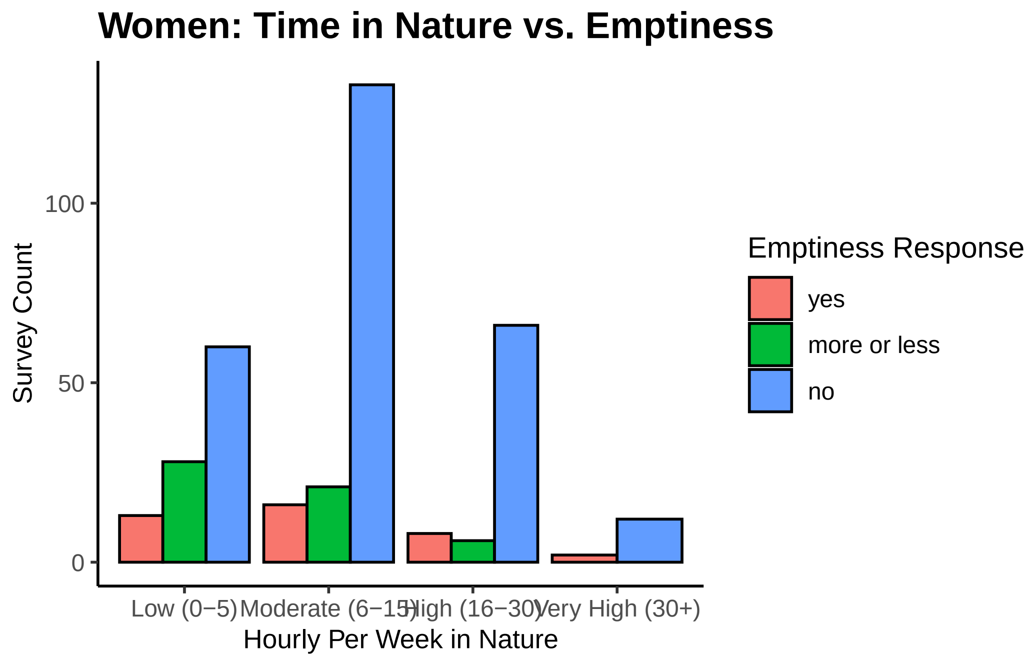
```
                  yes more or less  no
  Low (0-5)        13           28  60
  Moderate (6-15)  16           21 133
  High (16-30)      8            6  66
  Very High (30+)   2            0  12
```

```r
# Chi-Square Test
if (sum(table_women) == 0) {cat("No valid data in table for women")
  } else {
  chisq_test_women <- chisq.test(table_women)

cat("\n Chi-Square Test for Women:\n")
print(chisq_test_women)}
```

```
Warning in chisq.test(table_women): Chi-squared approximation may be incorrect
```

```

 Chi-Square Test for Women:

    Pearson's Chi-squared test

data:  table_women
X-squared = 21.973, df = 6, p-value = 0.001225
```

```r
# Bar chart
ggplot(Women_Data_Clean, aes(x = Nature_Hours_Group, fill = Lon_1)) +
  geom_bar(position = "dodge", color = "black") +
  labs(title = "Women: Time in Nature vs. Emptiness",
       x = "Hourly Per Week in Nature",
       y = "Survey Count",
       fill = "Emptiness Response") +
  theme_classic() +
  theme(plot.title = element_text(size = 14, face = "bold"),
    axis.title.x = element_text(size = 10),
    axis.title.y = element_text(size = 10),
    axis.text.x = element_text(size = 9),
    axis.text.y = element_text(size = 9))
```



Chi-Square Test of Independence Analysis #2: Women

A similar analysis was done for women to determine whether their time spent in nature is related to reported feelings of emptiness.

The test analysis yielded similar finding when compared to the previous test concerning men. Therefore, what was stated previously can also be stated here with women too.

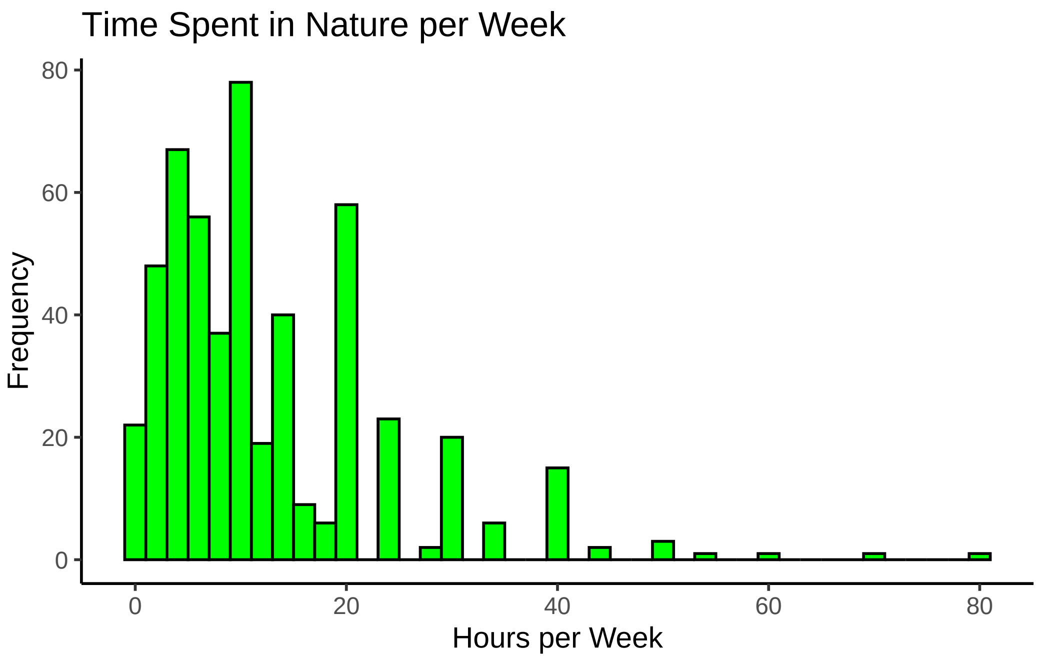
In essence, besides rejecting the null hypothesis in favor of the alternative one, these two test support the idea that spending time in nature may reduce feelings of emptiness. This is further evidence as the p-value was the same for woman (0.001) as it was for men.

6 ——————————Single T-Test——————————

```
# T-test
nature_hours_clean <- na.omit(Project_Data$D_hours)

t_test_result <- t.test(nature_hours_clean, mu = 5, alternative = "less")

print(t_test_result)
```

```

    One Sample t-test

data:  nature_hours_clean
t = 16.338, df = 514, p-value = 1
alternative hypothesis: true mean is less than 5
95 percent confidence interval:
     -Inf 13.71495
sample estimates:
mean of x
  12.9165
```

```
# Histogram
ggplot(Project_Data, aes(x = D_hours)) +
  geom_histogram(binwidth = 2, fill = "green", color = "black") +
  labs(title = "Time Spent in Nature per Week",
       x = "Hours per Week",
       y = "Frequency") +
      theme_classic()
```



Null Hypothesis: People spend 5 hours or more per week in nature

Alt Hypothesis: People spend less than 5 hours per week in nature

This is a single T-test to see whether people spend less than 5 hours per week in nature.

This histogram shows that people spend more than 5 hours in nature, with the mean being around 13 hours per week. The data is skewed to the right, meaning more people report higher time in nature.

The T-test gave a p-value of 1. It is not statistically significant to support the alternative hypothesis. It would in fact support the null hypothesis.

This also suggests that people, on average, are getting a decent amount exposure to nature. It is a good amount for mental well-being.

7 ——————————ANOVA ONE-WAY TEST——————————

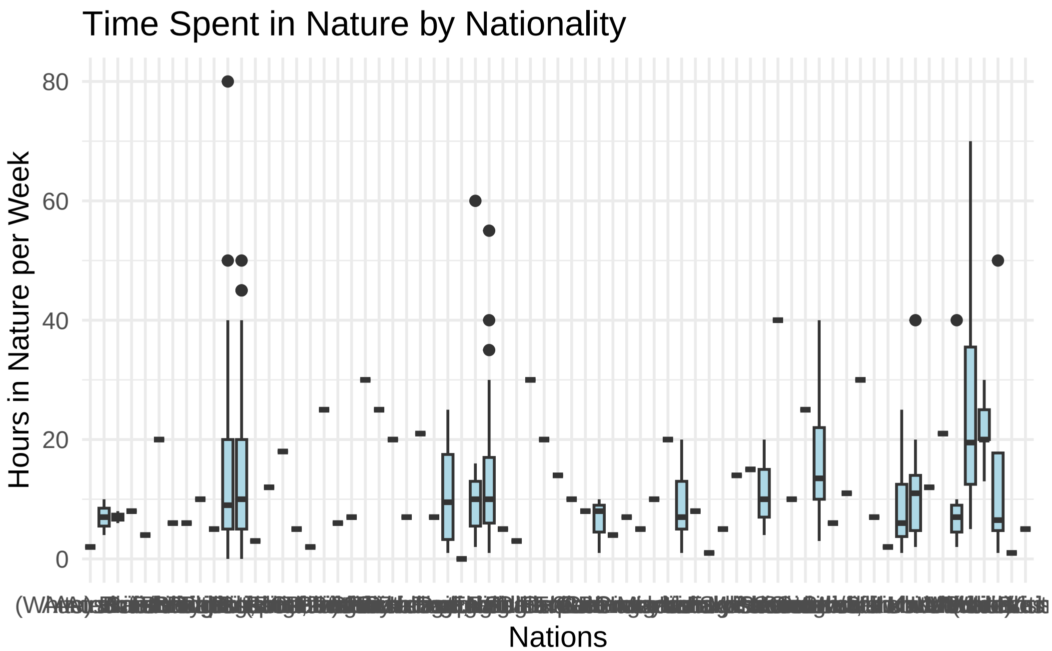
```
# Convert D_Nation to a factor
Project_Data$D_Nation <- as.factor(Project_Data$D_Nation)

# Remove any rows with missing values in D_Nation or D_hours
Project_Data_ANOVA <- subset(Project_Data, !is.na(D_Nation) & !is.na(D_hours))

# One-Way ANOVA
anova_nation <- aov(D_hours ~ D_Nation, data = Project_Data_ANOVA)

summary(anova_nation)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
D_Nation     68   6710   98.67   0.794  0.879
Residuals   446  55440  124.30
```

```
# Assumption Checks
leveneTest(D_hours ~ D_Nation, data = Project_Data_ANOVA)
```

```
Levene's Test for Homogeneity of Variance (center = median)
       Df F value Pr(>F)
group  68  0.7507 0.9273
      446
```

```r
# Box Plot
ggplot(Project_Data_ANOVA, aes(x = D_Nation, y = D_hours)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Time Spent in Nature by Nationality",
       x = "Nations",
       y = "Hours in Nature per Week") +
  theme_minimal()
```




```r
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
List of 1
 $ axis.text.x:List of 11
  ..$ family       : NULL
  ..$ face         : NULL
  ..$ colour       : NULL
  ..$ size         : NULL
  ..$ hjust        : num 1
  ..$ vjust        : NULL
  ..$ angle        : num 45
  ..$ lineheight   : NULL
  ..$ margin       : NULL
  ..$ debug        : NULL
  ..$ inherit.blank: logi FALSE
  ..- attr(*, "class")= chr [1:2] "element_text" "element"
 - attr(*, "class")= chr [1:2] "theme" "gg"
 - attr(*, "complete")= logi FALSE
 - attr(*, "validate")= logi TRUE
```